

Chemistry — class test

Chemistry (4CH1) · IGCSE · total 43 marks · target 40 · name: _____

Question 1 (1 marks · medium)

4CH1-S1-d

(a) 4CH1-1.20, 4CH1-1.21

[1]

The diagram shows part of the periodic table, with elements represented by the letters M, Q, R and T. The letters in the diagram represent elements but are not their chemical symbols.

Use the Periodic Table to help you answer this question.

																M	
	Q											R					
T																	

Which of these statements is true?

- A) T and Q form acidic oxides
- B) M conducts electricity
- C) M is a non-metal element
- D) R is a gaseous non-metal element

► Answer key — question 1

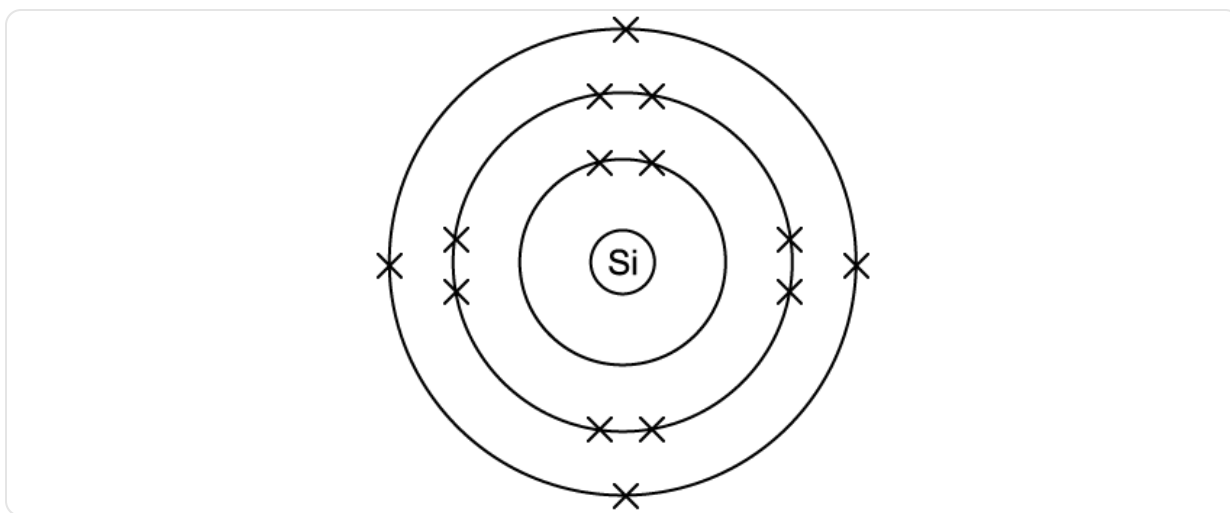
Question 2 (1 marks · medium)

4CH1-S1-d

(a) 4CH1-1.19, 4CH1-1.22

[1]

What is the electronic configuration of silicon?



- A) 2,4
- B) 2,2,4
- C) 2,2,8
- D) 2,8,4

► Answer key — question 2

Question 3 (1 marks · medium)

4CH1-S1-d

(a) 4CH1-1.19, 4CH1-1.22

[1]

Which diagram shows the electronic configuration of a nitrogen atom?

► Answer key — question 3

Question 4 (1 marks · medium)

4CH1-S1-d

(a) 4CH1-1.23, 4CH1-1.24

[1]

The following elements are found in group 0 of the periodic table.

He Ne Ar Kr Xe Rn

What explains their similar chemical and physical properties?

- A) They have the same number of valence electrons
- B) They are all gases
- C) They have the same number of protons and electrons
- D) They have full outer shells

► Answer key — question 4

Question 5 (1 marks · medium)

4CH1-S1-d

(a) 4CH1-1.19, 4CH1-1.22

[1]

Which element, T, Q, R, or M has the electronic configuration 2,8,3?

																M	
	Q											R					
T																	

- A) element M
- B) element Q
- C) element T
- D) element R

► Answer key — question 5

Question 6 (4 marks · hard)

4CH1-S1-d

(a) give 4CH1-1.18, 4CH1-1.24

[1]

The diagram shows part of the Periodic Table, with elements represented by the letters L, M, Q, R and T.

The letters in the diagram represent elements but are not their chemical symbols.

1 2		3 4 5 6 7 0					
L							
M							

Give the letter from the diagram that represents a noble gas.

source: 2019 paper Ju1C · Q2a

(b) state 4CH1-1.23

[1]

Elements L and M are in the same group.

State why they have similar chemical reactions.

source: 2019 paper Ju1C · Q2b

(c) explain 4CH1-1.18

[2]

An atom of element Q has 31 protons.

Use this information to explain how you can determine the number of protons in an atom of element R.

source: 2019 paper Ju1C · Q2c

► Answer key — question 6

Question 7 (4 marks · medium)

4CH1-S1-d

(a) complete 4CH1-1.18

[4]

The table contains statements about elements in the Periodic Table.

Complete the missing information.

[4]

Statement
The name of the element with atomic number 18.
The name of the element with relative atomic mass 7.
The name of the element in Group 3, Period 4.
The electronic configuration of the element phosphorus.

► Answer key — question 7

Question 8 (5 marks · hard)

4CH1-S1-d

(a)

[1]

In 1937, an airship full of hydrogen gas flew from Germany to America.



(Source: © Michael Rosskothen/Shutterstock)

Which property of hydrogen makes it a suitable gas to use in an airship?

- A) colourless
- B) insoluble in water
- C) low density
- D) no smell

source: 2019 paper Ju1C · Q5a

(b) explain 4CH1-1.24

[2]

Explain why helium is now used in airships instead of hydrogen.

source: 2019 paper Ju1C · Q5b

(c) multiple 4CH1-1.25, 4CH1-3.12

[2]

Hydrogen is used to manufacture ammonia, NH_3

Hydrogen is reacted with nitrogen using an iron catalyst.

i) Give a chemical equation for this reaction.

(1)

ii) State why a catalyst is used in this reaction.

(1)

source: 2019 paper Ju1C · Q5c

► Answer key — question 8

4CH1-S1-d

[4]

[illegible]

Each symbol may be used once, more than once or not at all.

(7)

(7)

(1)

(1)

source: 2020 paper Nov1CR · Q2a

[7]

source: 2020 paper Nov1CR · Q2b

► Answer key – question 9

Question 10 (6 marks · medium)

4CH1-S1-d

(a) name 4CH1-1.18

[3]

Use the Periodic Table to help you answer this question.

i) Name the element with atomic number 14

(1)

ii) Name the element with a relative atomic mass of 11

(1)

iii) Name the element in Group 2 and Period 3

(1)

source: 2021 paper Jun2C · Q1a

(b) multiple 4CH1-1.19

[3]

i) Determine the number of neutrons in a phosphorus atom with mass number 31

(1)

ii) State the electronic configuration of an aluminium atom.

(1)

iii) State why neon is unreactive.

(1)

source: 2021 paper Jun2C · Q1b

► Answer key — question 10

Question 11 (7 marks · easy)

4CH1-S1-d

(a) complete 4CH1-1.20, 4CH1-2.38, 4CH1-4.16

[5]

The box shows the names of some substances.

bromine	carbon dioxide	copper	iodine
methane	nitrogen	sulfur dioxide	water

Complete the table by choosing substances from the box that match the description.

Each substance may be used once, more than once or not at all.

Description	Substance
a good conductor of electricity	
an element that has a basic oxide	
a substance used as a fuel	
a major cause of acid rain	
a non-metallic element that is a solid at room temperature	

source: 2021 paper Ju1C · Q1a

(b) describe 4CH1-2.44

[2]

Describe a test for carbon dioxide.

source: 2021 paper Ju1C · Q1b

► Answer key — question 11

Question 12 (7 marks · easy)

4CH1-S1-d

(a) multiple 4CH1-1.18, 4CH1-1.19

[5]

This question is about chemical elements.

Use the Periodic Table to help you answer this question.

i) Identify the element with atomic number 5

(1)

ii) Give the symbol of a metallic element in Period 3

(1)

iii) Identify the element whose atoms contain 14 protons.

(1)

iv) Identify the element whose atoms have the electronic configuration 2.5

(1)

v) Give the name of the compound formed between oxygen and the element with atomic number 13

(1)

source: 2020 paper Nov1C · Q1a

(b) multiple 4CH1-1.23, 4CH1-1.24

[2]

The position of an element in the Periodic Table can be used to predict its properties.

i) Which group contains elements that are all unreactive?

(1)

☐	A	Group 2
☐	B	Group 5
☐	C	Group 6
☐	D	Group 0

ii) Which of these is the least reactive element in Group 1?

(1)

☐	A	caesium
☐	B	lithium
☐	C	potassium
☐	D	sodium

source: 2020 paper Nov1C · Q1b

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